NEUROSEG AI

AI-Powered Brain Tumor Detection and Segmentation from MRI

Graduation Project Documentation

Digital Egypt Pioneers Initiative (DEPI) — Phase Five

Team Information

Name	Role
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Project track: Artificial Intelligence / Deep Learning (confirm official track wording)

This document is a project-planning draft. Confirm the official project name, dataset, model, schedule and role assignments with the mentor and team before final submission.

Document Overview

This document covers the first three requested sections: Project Planning & Management, Literature Review, and Requirements Gathering. NeuroSeg AI is an educational AI prototype intended to segment suspected tumor regions in brain MRI scans; it is not a medical diagnostic device.

Project Summary

The system will take supported brain MRI data, apply documented preprocessing, run a deep-learning segmentation model, and produce a mask showing the predicted tumor location and approximate boundaries. A visualization layer will display the MRI and predicted mask overlay. Model performance will be assessed on held-out subjects using metrics such as Dice Similarity Coefficient and Intersection over Union (IoU).

Proposed baseline (to confirm): Python, PyTorch or TensorFlow, a U-Net-family segmentation model, and a permitted public dataset such as BraTS. Final choices depend on dataset terms, hardware and mentor guidance.

Contents

Section	Coverage
1. Project Planning & Management	Problem, objectives, scope, workflow, schedule, responsibilities, risks, success metrics.
2. Literature Review	Related approaches, comparison, limitations, research rationale, references.
3. Requirements Gathering	Stakeholders, functional and non-functional requirements, data/model requirements, acceptance criteria.

1. Project Planning & Management

1.1 Problem Statement

Manual identification and outlining of brain tumor regions on MRI scans can be time-consuming and may vary between reviewers. This project explores whether a deep-learning model can generate preliminary segmentation masks from labeled MRI data. Outputs are for educational and research demonstration only and must not replace assessment by

qualified medical professionals.

1.2 Project Objectives

- Prepare a documented pipeline for loading, validating, preprocessing and splitting labeled MRI data.
- Train and evaluate a segmentation model to identify suspected tumor regions.
- Visualize predicted masks over MRI images to make predicted location and boundaries easy to inspect.
- Evaluate with Dice and IoU, reporting per-region metrics when labels allow.
- Maintain a reproducible repository with code, setup instructions, evaluation results and permitted sample outputs.
- Document limitations and failure cases clearly.

1.3 Project Scope

In scope

- Use a permitted, labeled research dataset and document its version and terms.
- Split subjects/patients before deriving slices or patches to reduce data leakage.
- Train a baseline U-Net-family model; consider 2D or 3D input based on compute and data format.
- Generate segmentation masks, overlay visualizations and quantitative evaluation reports.
- Build a simple demo interface if time permits.

Out of scope

- Clinical diagnosis, treatment recommendations or autonomous patient-care decisions.
- Claims about hospitals, scanner types or populations absent from the test data.
- Tumor grading, prognosis or survival prediction unless separately scoped and validated.
- Deployment as a certified medical device.

1.4 Proposed Workflow

Dataset/license review → data inspection and subject-level split → preprocessing and augmentation → model training →

validation and model selection → held-out testing → mask overlay/demo → documentation and final review.

1.5 Illustrative Eight-Week Schedule

Adjust this example to the actual project start date and mentor plan. It is an implementation plan, not a statement that all work has already

been completed.

Week	Activities	Deliverable
1	Confirm scope, dataset access, related work, environment and repository.	Plan and repository skeleton
2	Inspect data/labels, confirm splits and preprocessing.	Data report and preprocessing pipeline
3	Implement baseline model and training loop.	Baseline run

4	Train, tune and track validation experiments.	Experiment logs and selected model
5	Test on held-out data; calculate metrics and inspect errors.	Evaluation report
6	Create overlay visualizations and demo interface.	Working demo
7	Test usability, fix issues and document limitations.	Release candidate
8	Finalize README, report and presentation.	Final package

1.6 Team Responsibilities

Member	Suggested responsibility
عبدالرحمن محمد عبدالحميد عبدالحكم (Team Leader)Team Leader)	Coordinate milestones, mentor communication, task board and final integration.
عبدالرحمن عادل عبدالحميد السيد	Dataset research, data inspection and preprocessing pipeline.

Member	Suggested responsibility
خالد حسين يوسف كامل	Model implementation, training and experiment tracking.
ايمن محمد فوزي عيسي	Evaluation metrics, error analysis and result reporting.
بسمله اسماعيل احمد كاظم	Demo interface, visualization, testing and README support.

These are proposed assignments based on typical project work, not confirmed commitments. The team should adjust them according to members' skills and mentor guidance.

1.7 Risk Management

Risk	Likelihood / Impact	Mitigation
Dataset access/license	Medium / High	Confirm terms early; use only permitted data and cite its source.
Limited GPU/time	High / Medium	Start with a baseline; consider patches or 2D slices if justified; record compute limits.
Class imbalance	High / High	Use suitable segmentation loss, augmentation and region-specific evaluation.
Train/test data leakage	Medium / High	Split by subject before generating slices/patches; keep test set untouched during tuning.
Overfitting / weak	Medium / High	Use validation, early stopping/regularization, and report limitations.

generalization

Misleading overlay Medium / High Test alignment and label predictions clearly; include a medical-use disclaimer.

Uneven workload or delays Medium / Medium Weekly check-ins, small tasks, clear owners and shared task board.

Privacy or secret exposure Low–Medium / High Use de-identified/public data; never commit patient data, passwords or tokens.

1.8 Success Metrics / KPIs

KPI	Measurement	Success rule
Segmentation overlap	Dice and IoU on held-out test data.	Report measured values; set targets after baseline validation.
Region-specific quality	Metrics per target class/region where available.	Do not let an aggregate score hide weak regions.
Reproducibility	Dependencies, configuration, split protocol and run instructions.	A teammate can reproduce the baseline from the README.
Demo usability	Test input, prediction, overlay and error paths.	Core demo checklist passes.
Delivery completeness	Code, docs, setup instructions and permitted outputs.	All agreed deliverables reviewed before submission.

For segmentation, Dice and IoU are generally more informative than plain pixel accuracy. Do not claim a performance target as achieved until it has been measured on the defined test set.

2. Literature Review

2.1 Review Objective

This review summarizes established methods relevant to brain MRI tumor segmentation. It supports selecting a U-Net-family baseline and a structured evaluation protocol. Results from different papers are not directly comparable unless data versions, splits, preprocessing, labels and evaluation procedures match.

2.2 Related Work

Study / method	Contribution	Strengths	Limitations / relevance
Menze et al. (2015), BRATS benchmark [1]	Introduced a multimodal MRI brain-tumor segmentation benchmark with expert annotations.	Foundational benchmark context; demonstrates segmentation challenges.	Older benchmark setup; versions and protocols have evolved. Not a current performance target.
Çiçek et al. (2016), 3D U-Net [2]	Extends U-Net to 3D volumetric segmentation.	Uses context across neighboring slices/volume.	Higher memory/compute needs; paper is not limited to brain tumors.
Isensee et al. (2021), nnU-Net [3]	Self-configuring biomedical segmentation pipeline.	Strong practical reference for preprocessing, training and post-processing choices.	Can be resource-intensive and more complex than a minimal student baseline.
Asthana & Hanmandlu (2022) [4]	Reviews/evaluates U-Net-based brain-tumor segmentation approaches on BraTS datasets.	Relevant comparison of U-Net approaches and tumor subregions.	Results depend on dataset version, labels and experimental protocol.

2.3 Comparison with NeuroSeg AI

Dimension	Existing approaches	Proposed project
Goal	Research segmentation methods and benchmark systems.	Educational end-to-end prototype with inspectable mask overlays.
Model	2D/3D U-Net, nnU-Net and enhanced variants.	Start with reproducible U-Net-family baseline; compare only if resources permit.
Evaluation	Dice and other region-specific metrics under paper-specific protocols.	Dice/IoU on held-out subject-level test set, plus error examples.
Output	Varies across papers and research implementations.	MRI with clearly labeled predicted mask overlay and setup documentation.
Contribution	New architectures, training strategies or benchmark results.	Reproducibility, clear visualization, honest limitations and full workflow.

2.4 Challenges Identified

- Tumor appearance, size, shape and boundaries vary across subjects and MRI modalities.
- Tumor regions may occupy a small fraction of image voxels, creating class imbalance.
- Different tumor subregions can have different segmentation difficulty.
- Performance may not generalize to scanners, institutions or populations absent from the training data.
- Visual overlays must be clearly marked as predictions and not confirmed diagnoses.

2.5 Research Rationale

NeuroSeg AI does not claim to invent a new architecture. Its practical value is a transparent, reproducible workflow

connecting data preparation, model training, quantitative evaluation and visualization, with documented failure cases

and limitations.

3. Requirements Gathering

3.1 Stakeholders and Needs

Stakeholder Needs / expectations

Project team Clear requirements, version control, reproducible setup, task ownership and

measurable milestones.

Mentor / evaluators Defined scope, justified method, reliable evaluation, citations and working

demonstration.

Researchers / students Understandable visual output, model limitations, dataset provenance and repeatable

experiments.

Potential clinical users Clearly labeled exploratory output and safeguards against treating it as clinical

(indirect) advice.

3.2 Functional Requirements

ID Requirement Priorit Acceptance criterion

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FR-01 Load supported MRI images or Must Valid input loads; invalid/corrupt files produce a clear

supported preprocessed examples. error.

FR-02 Apply the same preprocessing used Must Input shape and intensity handling follow documented

during training. settings.

FR-03 Generate a segmentation mask using trained model weights. Must Mask aligns with processed input and follows documented class labels.

FR-04 Display MRI and predicted mask overlay. Must User can distinguish scan from predicted region(s).

FR-05 Evaluate on labeled held-out test data. Must Dice/IoU reported with protocol and region definitions.

FR-06 Export visualizations and evaluation summaries. Should Exports do not expose sensitive patient data.

FR-07 Provide setup and sample-workflow instructions. Must A teammate can run the documented example in a clean setup.

FR-08 Display an educational-use disclaimer. Must Interface and README state that output is not a diagnosis.

3.3 Non-Functional Requirements

Category	Requirement	Verification
Performance	Reasonable inference time for documented input/hardware.	Record hardware, input size and average runtime.
Reliability	Handle invalid input and missing weights clearly.	Test valid, invalid and missing-file cases.
Usability	First-time user can follow main workflow.	Ask a teammate unfamiliar with code to run the demo.
Reproducibility	Document dependencies, configuration, split strategy and seeds where practical.	Fresh environment setup test.
Privacy/security	Do not expose identifiable patient data, secrets or tokens.	Review repository and .gitignore.
Maintainability	Modular code, meaningful names, README and setup steps.	Code review and setup check.
Portability	Document supported OS/Python environment and avoid local-only paths.	Run documented example in a clean environment where possible.

3.4 Data and Model Requirements

- Dataset: use a documented, properly licensed research dataset with expert-provided labels; BraTS is a candidate

subject to the selected edition's access terms.

- Data split: separate subjects/patients before deriving slices or patches; never tune against the final test set.
- Input definition: document MRI modalities, file format, orientation, spacing, resizing/cropping and normalization.
- Label definition: confirm exact class conventions for the chosen dataset version.
- Model: record architecture, loss, optimizer, training settings and checkpoint.
- Evaluation: report Dice and IoU per target class/region where possible and include qualitative error analysis.

3.5 Assumptions and Constraints

- The team can access a suitable public research dataset and comply with its license.
- The prototype is educational; clinical validation and regulatory approval are not assumed.
- Available hardware may limit full 3D training or the number of experiments.
- Final interface, dataset edition, MRI modalities and framework require team/mentor confirmation.
- Only public/de-identified data should be committed to the repository.

3.6 High-Level Acceptance Criteria

- A repeatable workflow maps supported MRI input to a predicted segmentation mask.
- Mask overlay is aligned and labeled as a model prediction.
- Held-out subject-level evaluation reports Dice and IoU with class definitions.
- Repository contains code, README, setup instructions, project documentation and permitted sample outputs.
- Limitations are documented and no clinical-diagnosis claim is made.

References

[1] B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS),” IEEE Transactions on

Medical Imaging, vol. 34, no. 10, pp. 1993–2024, 2015. DOI: 10.1109/TMI.2014.2377694.

https://pmc.ncbi.nlm.nih.gov/articles/PMC4833122/

[2] Ö. Çiçek et al., “3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation,” MICCAI, 2016, pp.

424–432. https://arxiv.org/abs/1606.06650

[3] F. Isensee et al., “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation,”

Nature Methods, vol. 18, pp. 203–211, 2021. DOI: 10.1038/s41592-020-01008-z.

https://doi.org/10.1038/s41592-020-01008-z

[4] P. Asthana and M. Hanmandlu, “U-Net-based semantic segmentation of brain tumors on BraTS datasets,”

International Journal of Imaging Systems and Technology, first published 13 April 2022. DOI: 10.1002/ima.22735.

https://onlinelibrary.wiley.com/doi/10.1002/ima.22735

Final Submission Checklist

- Verify spelling of the team members' names against the original Arabic names.
- Confirm the project name and official track wording.
- Confirm dataset edition/access terms, model framework, actual schedule and role assignments.
- Review the official Project Documentation Guidelines and required template.
- Upload this single PDF to the team's one GitHub repository and record the repository URL in the team's Google Drive

project sheet.

- Add README instructions; do not commit patient data, credentials, tokens or private files.

Medical-use disclaimer: NeuroSeg AI is a student research prototype. It is not intended to diagnose disease,

guide treatment, or replace a qualified medical professional. Independent validation is required before any

clinical use.